

WASP-52b

R-0489

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_180918 · created 2026-07-20T09:34:44+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2458379.8659 +/- 0.0037 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.215 +/- 0.039
Transit depth (Rp/Rs)^2	4.62 +/- 1.67 [%]
Orbital Inclination (inc)	85.53 +/- 1.97
Airmass coefficient 1 (a1)	0.957 +/- 0.018
Airmass coefficient 2 (a2)	0.041 +/- 0.04
Scatter in the residuals of the lightcurve fit is	1.89 %
Best Comparison Star	#3 - [388, 239]
Optimal Aperture	3.88
Optimal Annulus	7.99
Transit Duration (day)	0.078 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.215 ± 0.039 vs 0.1646 (deviation 1.29σ)
Duration fit vs published	0.078 d vs 0.0758 d (deviation 0.17σ)
Mid-time O–C	11.0 min
Depth SNR	5.5
Residual scatter	1.89 %

Light curve

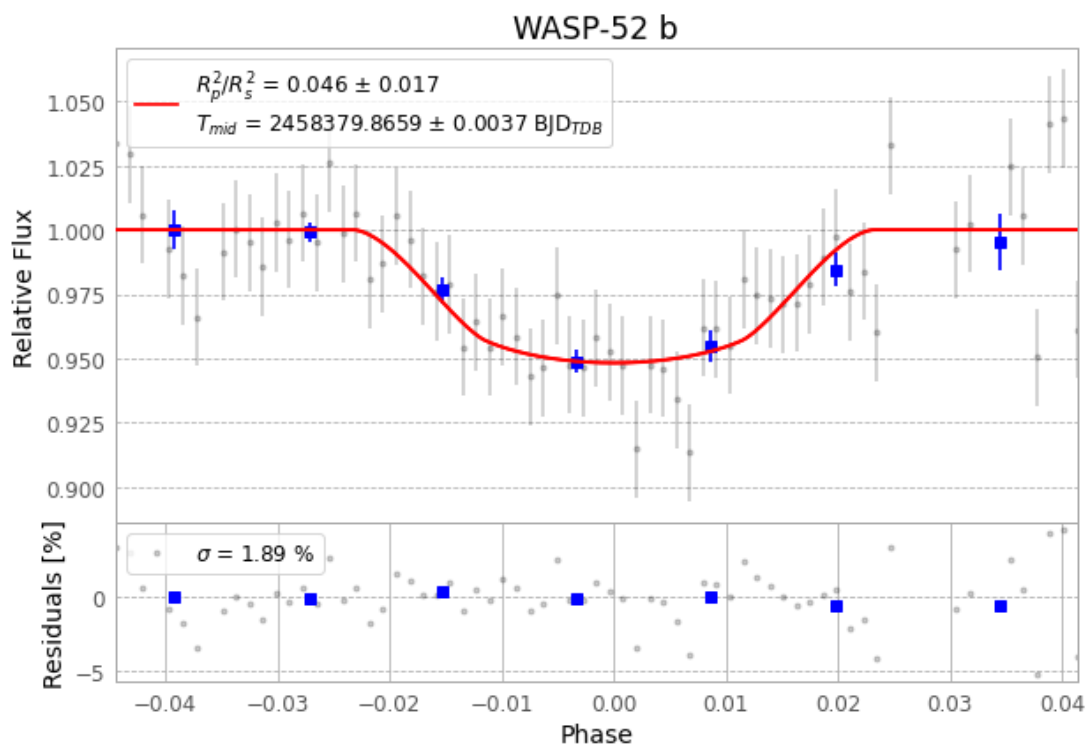


Figure 1 — FinalLightCurve_WASP-52 b_2018-09-17.png (SHA-256 46fcb2ada0fe74e2...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [278, 259], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8c9f3216cffa093f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

74 FITS frames (49 MB), WASP-52180918064512.FITS → WASP-52180918102712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-180918.log (bcc608884a93...), FinalLightCurve_WASP-52 b_2018-09-17.png (46fcb2ada0fe...), FinalLightCurve_WASP-52 b_2018-09-17.csv (6580486a209a...)

Compute resources

Wall time 335.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-20 09:34Z.